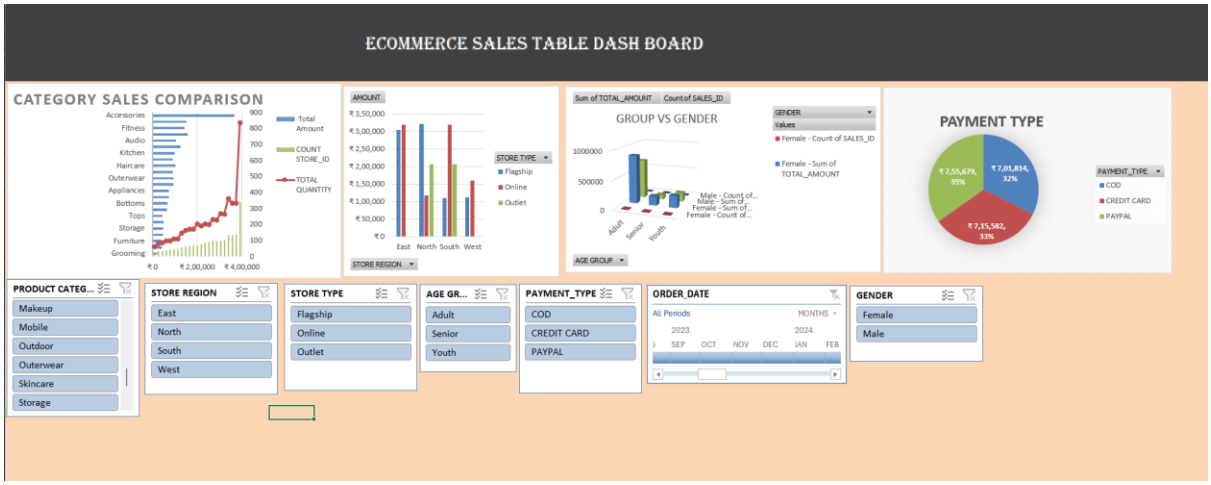


Customer_ID	Name	Age	Age Group	Gender	City	State	Country	Loyalty_Level
CUST1000	Sheena Bonilla	55	Adult	Male	Donaldton	North Carolina	USA	Unkown
CUST1001	Eugene Nelson	26	Adult	Female	Phillipsmouth	North Dakota	USA	Platinum
CUST1002	Dr. Benjamin Price	29	Adult	Female	Kylebury	Alabama	USA	Platinum
CUST1003	Jeremy Harris	59	Adult	Female	Thomastown	Ohio	USA	Bronze
CUST1004	Caroline Collins	64	Senior	Female	Nicholasshire	California	USA	Silver
CUST1005	Christine Vaughn	24	Youth	Male	East Amandamouth	Oregon	USA	Bronze
CUST1006	Amber Jackson	29	Adult	Female	Deborahborough	South Dakota	USA	Platinum
CUST1007	Daniel Boyd	29	Adult	Female	West Pamelafurt	West Virginia	USA	Bronze
CUST1008	Melissa Howard	53	Adult	Female	Maynardton	New Hampshire	USA	Gold
CUST1009	Ms. Maria Leach	64	Senior	Male	Melissashire	Pennsylvania	USA	Platinum
CUST1010	Dominic Page	53	Adult	Male	Port Michaelburgh	Pennsylvania	USA	Bronze
CUST1011	Brian Lopez	28	Adult	Female	East Krista	New York	USA	Gold
CUST1012	Kenneth Lambert	55	Adult	Female	North Kathryn	Kansas	USA	Platinum
CUST1013	Nathan Hernandez	32	Adult	Female	New Becky	Montana	USA	Silver
CUST1014	Ashley Cohen	53	Adult	Female	Lake Kristamouth	Nebraska	USA	Silver
CUST1015	Laura Alvarez	49	Adult	Female	Lawsonhaven	Washington	USA	Platinum
CUST1016	John Scott	24	Youth	Female	South Christopher	Pennsylvania	USA	Bronze
CUST1017	Adam Cruz	60	Senior	Female	Brandonville	Georgia	USA	Platinum
CUST1018	Rhonda Moss	43	Adult	Male	South William	Kansas	USA	Unkown
CUST1019	Michael Torres	46	Adult	Female	Gonzalezfort	New Jersey	USA	Platinum
CUST1020	Bridget Allen	59	Adult	Male	Lake Shaneburgh	Delaware	USA	Silver
CUST1021	Jeff Martinez	36	Adult	Female	Smithburgh	Oklahoma	USA	Gold
CUST1022	Paul Fox	33	Adult	Female	Michaelton	Hawaii	USA	Silver
CUST1023	Jonathan Steele	52	Adult	Female	Molinabury	Oregon	USA	Silver
CUST1024	Heather Collins	40	Adult	Male	Rogerschester	Idaho	USA	Platinum
CUST1025	Diana Taylor	59	Adult	Female	Vasqueztown	California	USA	Bronze
CUST1026	Amber Carlson	18	Youth	Female	East Michellemouth	Missouri	USA	Gold
CUST1027	Leslie Johnson	31	Adult	Male	Mollyshire	Utah	USA	Unkown
CUST1028	Joshua Ryan	27	Adult	Female	New David	New Jersey	USA	Platinum
CUST1029	David Arnold	54	Adult	Female	East Jennifer	California	USA	Silver
CUST1030	Jessica Acevedo	64	Senior	Female	Amybury	Pennsylvania	USA	Bronze

I USED TO CLEAN THE ENTIRE DATA WHILE USING FORMULA LIKE
PROPER(CLEAN(TRIM(B2)))
I Cleaned all the sheet in this work book

Fill the blank cell while using IF(Isblank) Function
IF(ISBLANK(H2),"unkown",H2)

Replace the inconsistency data with the help of Find and Replace
Repalced cusomer id

I Just Add Column of age Group help of IF Function
IF(C2<=25,"youth",IF(C2<=59,"ADULT","SENIOR"))

Product_ID	Product_Name	Category	Sub_Category	Brand	Cost	Stock
PROD0	Brother Laser Printer	Electronics	Computer	Hunt Inc	₹ 112	416
PROD1	Men'S Graphic T-Shirt	Clothing	Tops	Terry-Moore	₹ 343	481
PROD2	Women'S Casual Hoodie	Clothing	Outerwear	Sims Inc	₹ 255	352
PROD3	Men'S Running Jacket	Clothing	Outerwear	Pineda Ltd	₹ 365	452
PROD4	Women'S Yoga Pants	Clothing	Bottoms	Gonzales-Davis	₹ 306	487
PROD5	Men'S Leather Belt	Clothing	Accessories	Baldwin, Garza And Gonzalez	₹ 261	59
PROD6	Herbal Face Moisturizer	Beauty	Skincare	Davidson Plc	₹ 313	350
PROD7	Led Table Lamp	Home	Lighting	Harrell Llc	₹ 719	170
PROD8	Wireless Bluetooth Earbuds	Electronics	Audio	Perez-Johnston	₹ 439	142
PROD9	Adjustable Dumbbell Set	Sports	Fitness	Hill Plc	₹ 496	323
PROD10	Ceramic Kitchen Bowl Set	Home	Kitchen	Hamilton Group	₹ 299	308
PROD11	Hd Smart Tv Stick	Electronics	Accessories	Gibson, Hobbs And Weaver	₹ 449	271
PROD12	Hair Strengthening Shampoo	Beauty	Haircare	Rice Llc	₹ 116	460
PROD13	Usb-C Fast Charger	Electronics	Accessories	Dawson-Perkins	₹ 242	453
PROD14	Decorative Wall Mirror	Home	Decor	Luna, Anderson And Bowman	₹ 48	439
PROD15	4K Home Projector	Electronics	Appliances	Thomas Llc	₹ 465	301
PROD16	Beard Grooming Kit	Beauty	Grooming	Erickson-Green	₹ 337	256
PROD17	Soccer Training Cone Set	Sports	Accessories	Ford, Murphy And Wilson	₹ 145	367
PROD18	Women'S Denim Jeans	Clothing	Bottoms	Guerra-Hendrix	₹ 246	343
PROD19	Knee Support Brace	Sports	Equipment	May-Reese	₹ 21	65
PROD20	Outdoor Sports Water Bottle	Sports	Accessories	Soto-Ellis	₹ 132	264
PROD21	Multi-Purpose Storage Box	Home	Storage	Gardner-Montes	₹ 123	76
PROD22	Carbon Fiber Tennis Racket	Sports	Equipment	Flores, Pineda And Rodriguez	₹ 485	432
PROD23	Men'S Formal Shirt	Clothing	Tops	Moon, Mora And Santos	₹ 249	360
PROD24	Foldable Camping Chair	Sports	Outdoor	Cook Ltd	₹ 166	340
PROD25	Women'S Scarf Set	Clothing	Accessories	Brown And Sons	₹ 174	268
PROD26	Wireless Bluetooth Speaker	Electronics	Audio	Reed-Schmidt	₹ 322	213
PROD27	Decorative Floor Vase	Home	Decor	Khan, Jones And Paul	₹ 250	378
PROD28	Sports Stopwatch Timer	Sports	Accessories	Burns-Hansen	₹ 39	437
PROD29	Fabric Sofa Throw Blanket	Home	Decor	Hoffman Group	₹ 218	276
PROD30	Stainless Steel Cutlery Set	Home	Kitchen	Stevens, Fernandez And Martin	₹ 20	57

I USED TO FILL THE BLANK CELL IN THE COST AND STOCK COLUM AVERAGE FUNCTION
IF(ISBLANK(F2),AVERAGE(F2:F101),(F2))
IF(ISBLANK(G2),AVERAGE(G2:G101),G2)

HIGHLIGHTED STOCK >100 AND USED CONDITION FORMATING FOR COST COLUMN

Store_ID	Store_Name	Region	City	Store_Type
STORE0	Rivera-Warner	East	Washingtonfurt	Flagship
STORE1	Martinez, Adkins And Johnson	North	Danielmouth	Flagship
STORE2	Black-Horn	South	North Nancyview	Outlet
STORE3	Johnson-Wright	North	Hoodberg	Outlet
STORE4	Alexander Group	South	Port Sharon	Online
STORE5	Gray Ltd	North	New Michaelchester	Outlet
STORE6	Benson-Elliott	East	Port Josephberg	Online
STORE7	Martinez-Mullins	South	Williamsfurt	Online
STORE8	Bean, Jones And Park	West	Thompsonborough	Online
STORE9	Moore, Padilla And Butler	South	East Johnstad	Flagship
STORE10	Hill, Wagner And Ballard	East	East Davidville	Online
STORE11	Aguirre-Keller	South	Lake Michael	Online
STORE12	Thomas, Bridges And Harris	North	Sabrinatown	Online
STORE13	Gardner, Humphrey And Contreras	East	Johnsonville	Flagship
STORE14	Mcdonald Inc	South	Christopherland	Outlet
STORE15	Shelton Plc	North	East Jenniferstad	Flagship
STORE16	Frederick-Wright	West	New Jimmymouth	Flagship
STORE17	Holloway Ltd	North	Bensonhaven	Flagship
STORE18	Guerrero-Schmitt	East	East Stevechester	Flagship
STORE19	Mann-Austin	East	Vasquezborough	Online

I USED TO CLEAN THE ENTIRE DATA WHILE USING FORMULA LIKE
PROPER(CLEAN(TRIM()))
I Cleaned all the sheet in this work book

Sales_ID	Order_Date	Customer_ID	Product_ID	Store_ID	Quantity	Unit_Price	Discount	Payment_Type	Total_Amount
SALE0	07-07-2024	CUST1409	PROD97	STORE3	1	₹ 218	0.09	CREDIT CARD	₹ 198
SALE1	25-12-2023	CUST1467	PROD74	STORE17	3	₹ 104	0.27	PAYPAL	₹ 229
SALE2	03-06-2024	CUST1056	PROD49	STORE8	4	₹ 914	0.25	CREDIT CARD	₹ 2,742
SALE3	29-11-2023	CUST1346	PROD76	STORE8	1	₹ 922	0.16	COD	₹ 774
SALE4	06-01-2024	CUST1493	PROD59	STORE5	3	₹ 745	0.18	CREDIT CARD	₹ 1,833
SALE5	26-07-2024	CUST1116	PROD15	STORE7	3	₹ 147	0.19	COD	₹ 357
SALE6	12-07-2025	CUST1016	PROD56	STORE4	2	₹ 785	0.15	PAYPAL	₹ 1,335
SALE7	18-11-2023	CUST1152	PROD70	STORE8	4	₹ 342	0.23	PAYPAL	₹ 1,055
SALE8	04-05-2025	CUST1187	PROD99	STORE5	2	₹ 448	0.07	PAYPAL	₹ 834
SALE9	31-07-2025	CUST1464	PROD33	STORE0	2	₹ 873	0.23	PAYPAL	₹ 1,345
SALE10	13-02-2024	CUST1065	PROD24	STORE14	1	₹ 279	0.07	CREDIT CARD	₹ 259
SALE11	24-05-2025	CUST1320	PROD39	STORE1	4	₹ 287	0.17	CREDIT CARD	₹ 953
SALE12	24-08-2024	CUST1237	PROD88	STORE4	4	₹ 645	0.01	CREDIT CARD	₹ 2,553
SALE13	06-08-2025	CUST1443	PROD34	STORE16	2	₹ 317	0.30	PAYPAL	₹ 444
SALE14	21-06-2025	CUST1348	PROD24	STORE15	4	₹ 403	0.29	PAYPAL	₹ 1,146
SALE15	10-12-2024	CUST1435	PROD36	STORE9	2	₹ 871	0.04	COD	₹ 1,673
SALE16	13-09-2025	CUST1276	PROD65	STORE19	2	₹ 452	0.25	CREDIT CARD	₹ 678
SALE17	11-12-2024	CUST1079	PROD27	STORE4	2	₹ 503	0.23	CREDIT CARD	₹ 774
SALE18	13-08-2024	CUST1094	PROD39	STORE4	2	₹ 559	0.25	COD	₹ 839
SALE19	18-03-2024	CUST1239	PROD60	STORE13	4	₹ 514	0.04	COD	₹ 1,974
SALE20	05-07-2025	CUST1144	PROD72	STORE17	3	₹ 986	0.27	COD	₹ 2,159
SALE21	04-06-2024	CUST1396	PROD43	STORE18	1	₹ 590	0.22	CREDIT CARD	₹ 460
SALE22	12-04-2024	CUST1476	PROD69	STORE10	3	₹ 979	0.03	PAYPAL	₹ 2,850
SALE23	23-06-2024	CUST1368	PROD12	STORE8	1	₹ 719	0.08	CREDIT CARD	₹ 662
SALE24	08-09-2025	CUST1186	PROD67	STORE6	3	₹ 78	0.07	CREDIT CARD	₹ 217
SALE25	17-09-2024	CUST1381	PROD5	STORE3	4	₹ 853	0.14	PAYPAL	₹ 2,933
SALE26	03-01-2024	CUST1213	PROD98	STORE8	4	₹ 156	0.15	CREDIT CARD	₹ 532
SALE27	22-02-2025	CUST1200	PROD77	STORE8	3	₹ 815	0.17	CREDIT CARD	₹ 2,031
SALE28	23-09-2025	CUST1148	PROD13	STORE2	2	₹ 74	0.11	PAYPAL	₹ 132
SALE29	11-01-2024	CUST1348	PROD99	STORE3	2	₹ 452	0.15	CREDIT CARD	₹ 768
SALE30	18-10-2023	CUST1028	PROD5	STORE19	1	₹ 429	0.13	PAYPAL	₹ 374

No duplicate Found In Sales ID Column

I Used Mode Function For Quantity Column
IF(ISBLANK(F2),MODE..MULT(F2:F2001),F2)

I USED AVERAGE FUNCTION TO FILL BLANK IN UNTI PRICE
IF(ISBLANK(G2),AVERAGE(G2:G2001),G2)

FOR TOTAL AMOUNT CALCULATION
Total Amount = quantity * Unit price *(1-Discount)

Descriptive Statistics Analysis

UNIT PRICE	
Mean	513.9831146
Standard Error	6.307795632
Median	522.43
Mode	871.34
Standard Deviation	282.0931964
Sample Variance	79576.57147
Kurtosis	-1.189707573
Skewness	-0.036751141
Range	979.8
Minimum	20.09
Maximum	999.89
Sum	1027966
Count	2000
Largest(1)	999.89
Smallest(1)	20.09
Confidence Level(95.0%)	12.37054236

TOTAL AMOUNT	
Mean	1086.537817
Standard Error	18.30058503
Median	868.7182
Mode	#N/A
Standard Deviation	818.4270432
Sample Variance	669822.8251
Kurtosis	0.05944133
Skewness	0.88192118
Range	3889.7838
Minimum	18.459
Maximum	3908.2428
Sum	2173075.634
Count	2000
Largest(1)	3908.2428
Smallest(1)	18.459
Confidence Level(95.0%)	35.89021833

V & x look up Formulas

COMBAIN CUSTOMER NAME
VLOOKUP(C2,Customer_Dim!$A$2:$I$501,2,FALSE)
COMBAIN CUSTOMER CITY
VLOOKUP(C2,Customer_Dim!$A$2:$I$501,6,FALSE)
COMBAIN CUSTOMER AGE
VLOOKUP(C2,Customer_Dim!$A$2:$I$501,3,FALSE)
COMBAIN CUSTOMER GENDER
VLOOKUP(C2,Customer_Dim!$A$2:$I$501,4,FALSE)
COMBAIN PRODUCT NAME
VLOOKUP(I2,Product_Dim!$A$2:$G$101,2,FALSE)
COMBAIN CATEGORY
VLOOKUP(I2,Product_Dim!$A$2:$G$101,4,FALSE)
COMBAIN BRAND
VLOOKUP(I2,Product_Dim!$A$2:$G$101,5,FALSE)
I USED X LOOKUP FORMUL FOR COMBAIN THE STORE DATA
XLOOKUP(M2,Store_Dim!$A$2:$A$21,Store_Dim!$B$2:$B$21,"NOT FOUND")
XLOOKUP(M2,Store_Dim!$A$2:$A$21,Store_Dim!$C$2:$C$21,"NOT FOUND")
XLOOKUP(M2,Store_Dim!$A$2:$A$21,Store_Dim!$E$2:$E$21,"NOT FOUND")

- ✓ **Category vs Sales**

Descriptive – What happened?

- Accessories had the **highest sales – ₹3,66,401.82**.
- Grooming had the **lowest sales – ₹21,748.59**.
- Accessories also had the highest quantity sold (**837 units**).

Diagnostic – Why / What is related?

- Accessories sales are high because more units were sold.
- Higher quantity is linked with higher sales.

Predictive – What may happen?

- Accessories may continue to have high sales if the same trend continues.
- Grooming may continue to have lower sales.

Prescriptive – What should we do?

- Keep enough stock for Accessories.
- Promote low-selling categories like Grooming.
- Check why some categories have low sales.

- ✓ **Region vs Store Type**

Descriptive

- **North** had the highest sales – **₹6,43,578.69**.
- **West** had the lowest sales – **₹2,70,609.03**.
- **Online** had the highest sales among store types – **₹9,14,963.12**.

Diagnostic

- North has strong Flagship and Outlet sales.
- South has strong Online sales.
- West has lower overall sales.

Predictive

- North and South may continue to perform well.
- Online sales may continue to be important.

Prescriptive

- Improve sales in West.
- Continue focusing on Online sales.

- Check which store type works best in each region.

✓ **Age Group vs Gender**

Descriptive

- **Adults** had the highest sales – **₹15,65,705.27**.
- Female sales – **₹12,31,593.68**.
- Male sales – **₹9,41,481.95**.

Diagnostic

- Adults have the highest sales because they have the highest number of orders.
- Female customers generated more sales than male customers.

Predictive

- Adults may continue to be the main customer group.
- Female customers may continue to contribute more sales.

Prescriptive

- Focus marketing on Adult customers.
- Create targeted offers for Youth and Senior customers.
- Study products preferred by female and male customers.

✓ **Payment Type**

Descriptive

- **PayPal** had the highest sales – **₹7,55,679.41**.
- Credit Card – **₹7,15,582.14**.
- COD – **₹7,01,814.08**.

Diagnostic

- PayPal has the highest sales and highest number of orders.
- All three payment methods are being used regularly.

Predictive

- PayPal may continue to be the leading payment method.

Prescriptive

- Keep all three payment options available.
- Promote the payment methods customers prefer.
- Check if payment method affects customer orders.